Lab 4.4 source pack

All records are fictional training data. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use outputs to operate, maintain or approve real equipment.

research-question.txt

Compare ultrasonic and electromagnetic flow measurement for a fictional conductive-water application. Identify operating principles, installation constraints and evidence gaps. Use accessible manufacturer technical pages or recognised standards publisher pages. Distinguish full standard clauses from catalogue summaries; do not invent clause numbers or claim conformity. Provide title, publisher, date/version, URL and supported claim for every source. Explain which missing application conditions prevent selection.

process-data.csv

sample_id: 1; fill_L: 9.9; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 2; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 3; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 4; fill_L: 10.2; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 5; fill_L: 9.8; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 6; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 7; fill_L: 9.9; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 8; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 9; fill_L: 10.2; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 10; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 11; fill_L: 9.9; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 12; fill_L: 9.8; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 13; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 14; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 15; fill_L: 10.2; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 16; fill_L: 9.9; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 17; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 18; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 19; fill_L: 9.8; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 20; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 21; fill_L: 10.2; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 22; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 23; fill_L: 9.9; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 24; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 25; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 26; fill_L: 9.8; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 27; fill_L: 10.0; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 28; fill_L: 10.2; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 29; fill_L: 9.9; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 30; fill_L: 10.1; phase: baseline; LSL_L: 9.4; USL_L: 10.6

sample_id: 31; fill_L: 10.0; phase: monitoring; LSL_L: 9.4; USL_L: 10.6

sample_id: 32; fill_L: 10.2; phase: monitoring; LSL_L: 9.4; USL_L: 10.6

sample_id: 33; fill_L: 10.8; phase: monitoring; LSL_L: 9.4; USL_L: 10.6

sample_id: 34; fill_L: 10.1; phase: monitoring; LSL_L: 9.4; USL_L: 10.6

sample_id: 35; fill_L: 9.9; phase: monitoring; LSL_L: 9.4; USL_L: 10.6

defects.csv

category: Calibration evidence; count: 12

category: Blocked strainer; count: 8

category: Air ingress; count: 5

category: Other; count: 5

claim-register.csv

claim: Nominal flow is 50 m3/h; source: DES-01; version: rev2; clause_or_page: 2; verified_by: ; status: Pending

claim: Material is confirmed; source: No source; version: ; clause_or_page: ; verified_by: ; status: Unsupported